Nupura Kane

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SKILLS SUMMARY:

Tools: Microsoft Office, PSS/E, TARA Powerflow, MATLAB, Power World, ASPEN, Microsoft Visio.
Interconnection Studies, Transmission Studies,Technical Reports, Steady State Analysis, Stability Analysis, Power Systems.

EDUCATION:

Arizona State University, Tempe, AZ (GPA: 3.5/4) Aug '21-May '23
Master of Science – Electrical Engineering (Power and Energy Systems): Resilient Smart Electric Grids, Power System Operations and Planning, Electric Energy Markets, Power Transmission and Distribution, Sustainable Energy Technology, Wide Area Monitoring Systems (WAMS) Applications in Power Systems

Savitribai Phule Pune University, Pune, India Aug '16-Nov '20
Bachelor of engineering – Electrical Engineering: Electrical Machines, PLC and SCADA, Energy Audit and Management, Renewable Energy, Control Systems, Power Systems, Smart Grids.

WORK EXPERIENCE:

Mitsubishi Electrical Power Products Inc Jun '23-Present
System Studies Engineer II Aug'24-Present

- Steady state analysis for ERCOT interconnection projects
- Detailed technical reports including the analysis results and overloads.
- Project modeling and post case analysis for renewable energy projects.

System Studies Engineer I Jun '23-Aug'24

- Perform multiple feasibility studies for new projects in the ISO interconnection queue.
- Prepare secure base cases for the project scope for different load levels as per transmission planning guidelines.
- Model Post Project layout for projects, including offshore wind, battery energy storage.
- Conduct Steady State Analyses in PSS/E, including N-1, N-1-1 and Short Circuit.
- Identify line and system upgrades, based on the upgraded contingencies.
- Draw project breaker diagrams in Visio for study reports.
- Analyse branch loadings and voltage violations in TARA for pre and post cases of the project.
- Prepare comprehensive technical reports including solutions and changes suggested mitigate project impact.
- Updating subsequent report iterations based on the customer's additions and suggestions.

Power System Engineering Division Intern Feb '23-Apr '23

- Conducted feasibility studies of new interconnections in existing grids in queue. May '22-Aug '22
- Performed load flow analysis, creating updated sliders.
- Data review of new battery, solar, or offshore wind project systems.
- Batch Run and system runs for different dispatch cases in TARA.
- Conducted Short Circuit Analysis in ASPEN and formulating the results as per the report standards.

Arizona State University, Graduate Teaching Assistant, Tempe, AZ Aug '22-Dec '22

- Teaching Assistant for 'Power Systems and Electronics'
- Set up and conducted weekly lab sessions and office hours.
- provided academic assistance for basic concepts of power systems, motors, and electronics.

PROJECTS:

Impact of Forest Fires on Transmission lines: This project was a review paper which focused on the causes of forest fire damaging the transmission lines, which if not controlled properly, can damage the grid and lead to blackouts. Various solutions like optimized power shutoff in the grid and prevention were suggested to protect the lines.

Selection of Circuit Breakers for fault protection: Fault analysis for a three phase, single line-to-ground, line-to-line, and double line-to-ground) on a Metropolis Light and Power System performed using PowerWorld. Appropriate ratings for the circuit breakers were studied and suggested for system protection.